WORLD SUSTAINABILITY

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INTRODUCTION

This project demonstrates a full-stack data pipeline designed to analyze 19 years of global sustainability metrics across 173 nations. By architecting a relational database in MySQL and synthesizing complex indicators in Power BI, this study transforms raw World Bank data into actionable insights. The objective was to move beyond static reporting to identify the socio-economic drivers behind the UN Sustainable Development Goals (SDGs)

An End-to-End Data Pipeline utilizing Excel, MySQL, and Power BI

End to End Process

01

ETL & Pre-processing (Excel) Standardized 54 variables into a clean snake_case format and addressed data sparsity through meticulous missing-value imputation.

02

Relational Architecture (MySQL) Decomposed flat data into an optimized Star Schema (Fact and Dimension tables), enforcing data integrity through Foreign Keys and Composite Primary Keys.

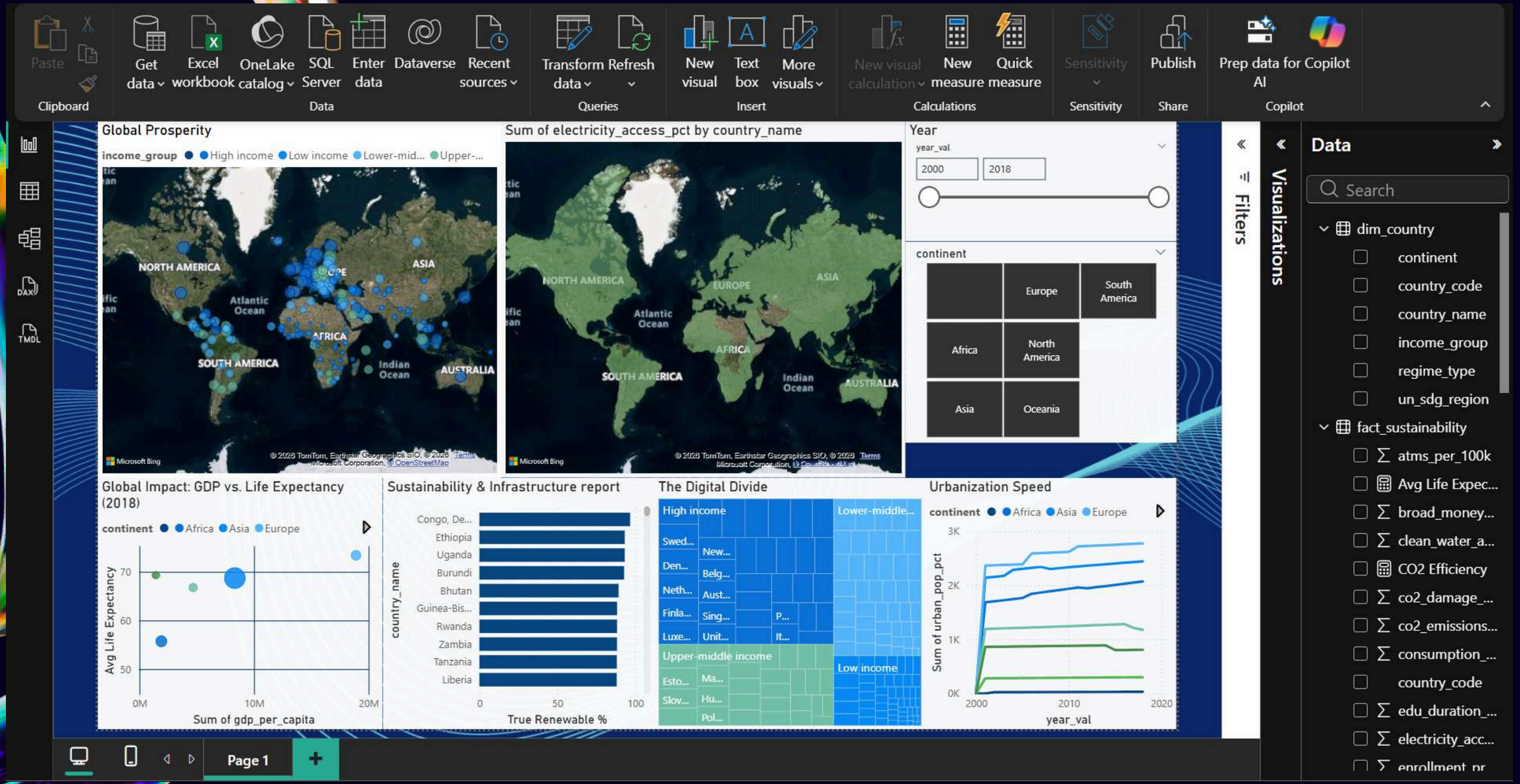
03

Analytical Engineering (MySQL) Developed complex scripts utilizing Joins and Aggregations to extract 10 strategic insights, ranging from Carbon Efficiency to the Digital Divide.

04

Intelligence Visualization (Power BI) Authored advanced DAX Measures to drive an interactive, 3-page dashboard, translating raw logic into high-fidelity Geospatial and Time-Series stories.

Power BI Dashboard



Question Answered

To extract meaningful intelligence from the 50+ variables in this dataset, I developed a focused analytical framework. These ten research questions were designed to explore the intersection of economic growth, social equity, and environmental sustainability, moving from high-level global trends to granular country-level disparities

01. Does a higher GDP per capita guarantee a longer life expectancy?

02. Which 10 countries have the highest percentage of renewable energy consumption?

03. Which countries produce the most GDP for every unit of CO2 emitted?

04. Is there a correlation between internet access and a country's income classification?

05. Which countries have the most crowded classrooms?

06. Which continents have the highest average percentage of women in national parliaments ?

07. Which countries have seen the fastest shift from rural to urban living over the 19-year period?

08. How many people (total population) still live in countries where electricity access is below 50%?

09. In which countries is the female unemployment rate significantly higher (e.g., >5% difference) than the male rate?

10. Do "Full Democracies" have higher renewable energy output than other regime types?

Does a higher GDP per capita guarantee a longer life expectancy?

```sql
-- Q1: Does a higher GDP per capita guarantee a longer life expectancy?

SELECT dim_country.country_name, fact_sustainability.gdp_per_capita, fact_sustainability.life_expectancy
FROM fact_sustainability JOIN dim_country
ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
ORDER BY fact_sustainability.gdp_per_capita DESC;

-- Key Insight:
-- You will notice some "Upper-Middle Income" countries have higher
-- life expectancies than "High Income" countries with double their GDP,
-- suggesting that social safety nets are as important as raw wealth.

-- The Findings:
-- While a higher gdp_per_capita generally correlates with a higher life_expectancy,
-- the relationship is not a straight line. Many "High Income" countries plateau around 80-83 years.
```

Result Grid | Filter Rows:

country_name	gdp_per_capita	life_expectancy
Luxembourg	116597.2956	82.29512195
Switzerland	86429.50082	83.75365854
Norway	82267.80932	82.75853659
Ireland	79297.7292	82.25609756
Iceland	74348.36228	82.86097561
Singapore	66679.04649	83.29756098
Qatar	65907.94843	80.1
United States	63064.41841	78.63902439
Denmark	61598.5367	80.95365854

Which 10 countries have the highest percentage of renewable energy consumption?

```sql
-- Q2: Which 10 countries have the highest percentage of renewable energy consumption?

SELECT dim_country.country_name, fact_sustainability.renewable_energy_consumption_pct
FROM fact_sustainability JOIN dim_country ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
ORDER BY fact_sustainability.renewable_energy_consumption_pct DESC LIMIT 10;

-- The Findings:
-- Many of the top 10 countries are actually in Sub-Saharan Africa.

-- Key Insight:
-- In developing nations, this is often due to Hydropower or a lack of fossil fuel infrastructure.
```

Result Grid | Filter Rows: | Export

country_name	renewable_energy_consumption_pct
Congo, Dem. Rep.	96.38369751
Uganda	90.33390045
Ethiopia	89.92310333
Gabon	89.88710022
Liberia	87.21330261
Guinea-Bissau	86.78929901
Rwanda	85.69450378
Burundi	85.52690125
Chad	85.25779724
Zambia	85.104599

Which countries produce the most GDP for every unit of CO2 emitted?

```sql
SELECT
dim_country.country_name,
(fact_sustainability.gdp_total_usd / fact_sustainability.co2_emissions_mt)
AS gdp_per_co2_unit
FROM fact_sustainability
JOIN dim_country ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018 AND fact_sustainability.co2_emissions_mt > 0
ORDER BY gdp_per_co2_unit DESC;
```

Result Grid | Filter Rows:

country_name	gdp_per_co2_unit
Congo, Dem. Rep.	20991188800.089645
Switzerland	19948502507.114784
Sweden	13288320643.585691
Chad	11149967309.52381
Uruguay	10319104009.59693
Denmark	10302732965.85957
Norway	9973070427.678123
Ireland	9947684457.387316
Singapore	9820309235.269537

Is there a correlation between internet access and a country's income classification?

```sql
SELECT dim_country.income_group, AVG(fact_sustainability.internet_users_pct) AS avg_internet_access
FROM fact_sustainability JOIN dim_country ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
GROUP BY dim_country.income_group
ORDER BY dim_country.income_group;
```

Result Grid | Filter Rows:

income_group	avg_internet_access
	0
High income	78.3396234569697
Low income	22.673457159818177
Lower-middle income	41.093328037111114
Upper-middle income	59.501476849642856

Which countries have the most crowded classrooms?

```sql
SELECT dim_country.country_name, fact_sustainability.pupil_teacher_ratio
FROM fact_sustainability
JOIN dim_country
ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
ORDER BY fact_sustainability.pupil_teacher_ratio DESC
LIMIT 10;

-- The Findings:
-- Sub-Saharan Africa and parts of Southern Asia often show ratios exceeding 40:1, while Europe averages 12:1.

-- Key Insight:
-- High ratios are a Leading Indicator of future economic struggle.
-- If teachers are overwhelmed, the quality of "Human Capital" decreases, slowing down long-term sustainability.
```

Result Grid Filter Rows:

country_name	pupil_teacher_ratio
Rwanda	59.50859
Malawi	58.67919
Mozambique	55.27379
Tanzania	50.63304
Cameroon	44.83412
Pakistan	44.0773
Burundi	42.52383
Cote d'Ivoire	41.8223
Cambodia	41.7035
Togo	40.14504

Which continents have the highest average percentage of women in national parliaments?

```sql
SELECT dim_country.continent, AVG(fact_sustainability.women_parliament_pct) AS avg_women_seats
FROM fact_sustainability
JOIN dim_country
ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
GROUP BY dim_country.continent
ORDER BY avg_women_seats DESC;
```

Result Grid | Filter Rows:

continent	avg_women_seats
	33.84615385
Europe	28.633076061282054
South America	27.811932905454544
North America	23.559561746368427
Africa	20.962298565499996
Asia	16.634810507075
Oceania	16.009344479

Which countries have seen the fastest shift from rural to urban living over the 19-year period?

```sql
SELECT
    dim_country.country_name,
    (MAX(fact_sustainability.urban_pop_pct) - MIN(fact_sustainability.urban_pop_pct)) AS urbanization_growth
FROM
    fact_sustainability
        JOIN
    dim_country ON fact_sustainability.country_code = dim_country.country_code
GROUP BY dim_country.country_name
ORDER BY urbanization_growth DESC
LIMIT 10;

-- The Findings:
-- The fastest urbanization is happening in Asia and Africa.

-- Key Insight:
-- increasing "gdp_per_capita" = increasing "co2_emmision". "SUSTAINAIBLITY CONCERNS !!!"
```

Result Grid | Filter Rows:

country_name	urbanization_growth
Kuwait	100
Singapore	100
Qatar	99.135
Belgium	98.001
Uruguay	95.334
Israel	92.418
Argentina	91.87
Japan	91.616
Netherlands	91.49
Luxembourg	90.981

How many people (total population) still live in countries where electricity access is below 50%?

```sql
SELECT SUM(fact_sustainability.total_population)
FROM fact_sustainability
WHERE  fact_sustainability.electricity_access_pct < 50
AND fact_sustainability.year_val = 2018;

-- The Findings:
-- Hundreds of millions of people still lack basic electricity access, primarily concentrated in rural areas.

-- Key Insight:
-- This finding highlights the Rural-Urban Divide.
-- Even in countries with 100% urban electricity, rural access can be as low as 30%.
```

Result Grid | Filter Rows:

	SUM(fact_sustainability.total_population)
▶	563360224

In which countries is the female unemployment rate significantly higher (e.g., >5% difference) than the male rate?

```sql
SELECT
dim_country.country_name,
fact_sustainability.unemployment_female_pct,
fact_sustainability.unemployment_male_pct,
(fact_sustainability.unemployment_female_pct - fact_sustainability.unemployment_male_pct)
AS gender_unemployment_gap
FROM fact_sustainability JOIN dim_country ON
fact_sustainability.country_code =  dim_country.country_code
WHERE fact_sustainability.year_val = 2018
AND (fact_sustainability.unemployment_female_pct - fact_sustainability.unemployment_male_pct) > 5
ORDER BY gender_unemployment_gap DESC;

-- The Findings:
-- In many regions, female unemployment is significantly higher,
-- but in some developed nations, the gap is nearly zero.

-- Key Insight: A high gap (where female unemployment > male) often correlates with a lower Women Business and the Law Index Score,
-- showing that legal barriers prevent women from entering the workforce.
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

country_name	unemployment_female_pct	unemployment_male_pct	gender_unemployment_gap
Saudi Arabia	22.6	2.9	19.700000000000003
Egypt, Arab Rep.	21.3	6.8	14.5
Tunisia	22.8	12.5	10.3
Greece	24.2	15.4	8.799999999999999
Iran, Islamic Rep.	19	10.5	8.5
Jordan	24.1	17	7.100000000000001
Oman	7.1	0.8	6.3
Dominican Republic	9	3.7	5.3
Mauritius	9.5	4.4	5.1

Do "Full Democracies" have higher renewable energy output than other regime types?

```sql
SELECT
    dim_country.regime_type,
    AVG(fact_sustainability.renewable_electricity_pct) AS avg_renewable_output
FROM fact_sustainability
JOIN dim_country ON fact_sustainability.country_code = dim_country.country_code
WHERE fact_sustainability.year_val = 2018
GROUP BY dim_country.regime_type
ORDER BY avg_renewable_output DESC;

-- The Findings:
-- "Electoral Democracies" typically have higher renewable_electricity_pct output
-- and better environmental regulations.
```

Result Grid | Filter Rows:

regime_type	avg_renewable_output
Electoral Democracy	0.37036958130434783

The End

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